

Western Mathematics

2020

YEAR 11
FINAL
EXAMINATION

Mathematics Standard

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks : **Section I – 10 marks** (pages 2 – 6)
80

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 70 marks (pages 7 – 29)

- Attempt Questions 11 – 36
- Allow about 1 hour and 45 minutes for this section

Section I

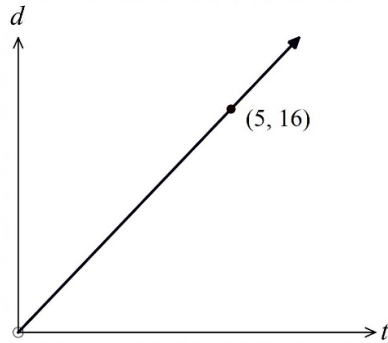
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

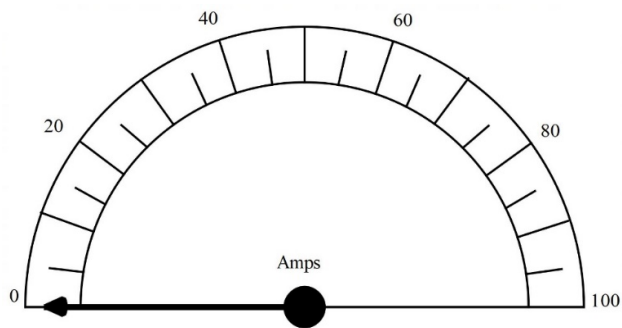
Use the multiple-choice answer sheet for Questions 1 – 10

1. Two variables t and d vary directly, so that $d = kt$.
The graph below shows the relationship.



What is the value of k ?

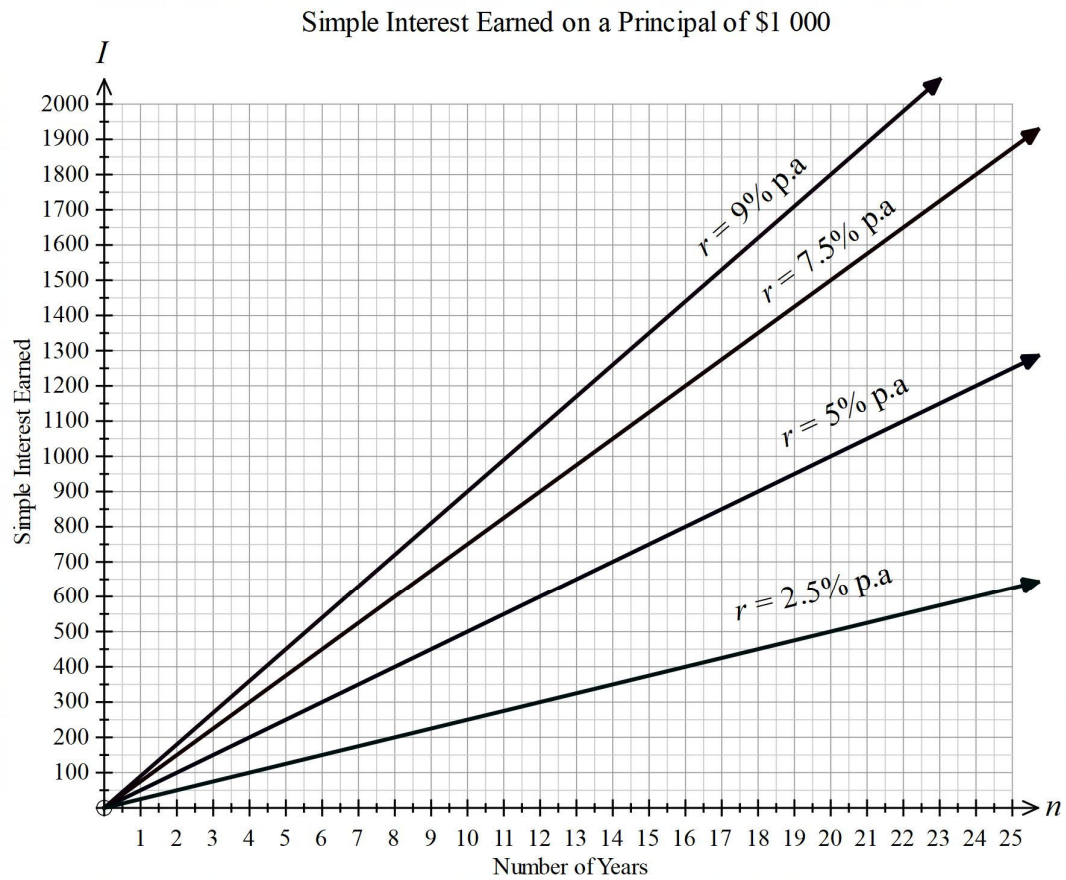
- (A) 0.3125
(B) 1.6
(C) 3.2
(D) 11
2. A meter for measuring electric current in amps is shown below.



An electrician records a reading of 45 amps using this meter.
Between what values might the actual amount of current lie?

- (A) 35 amps and 55 amps
(B) 40 amps and 50 amps
(C) 42.5 amps and 47.5 amps
(D) 44.5 amps and 45.5 amps

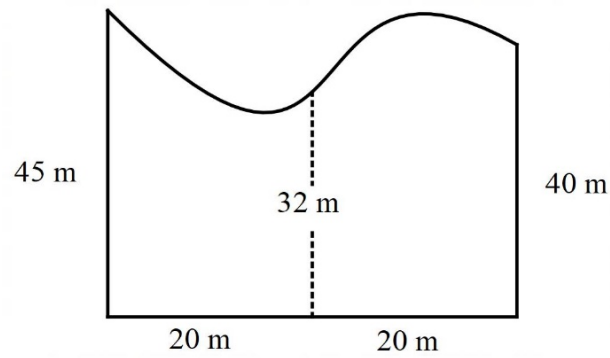
3. Jeremy has an amount of money ($\$X$) to invest at simple interest rates. The graph below compares the interest earned at various interest rates on a principal of \$1 000.



Using the graphs above, which calculation could be used to estimate how much interest he would earn if he invested his money at 9% p.a. for 20 years?

- (A) $\$ \frac{1.8}{X}$
- (B) $\$1.8X$
- (C) $\$ \frac{1800}{X}$
- (D) $\$1800X$

4. The measurements below are taken of a paddock which has an irregular boundary.



Estimate its area using two applications of the trapezoidal rule.

- (A) $1\,280\text{ m}^2$
(B) $1\,490\text{ m}^2$
(C) $1\,560\text{ m}^2$
(D) $1\,700\text{ m}^2$
5. Leigh is paid \$54.00 per hour for a normal 38-hour week, with overtime at time and a half for any hours over that.

What would Leigh earn when she worked a 42-hour week?

- (A) \$2\,376.00
(B) \$2\,889.00
(C) \$3\,402.00
(D) \$3\,564.00

6. Miriam is a health worker who assesses patients entering a hospital.

One of the questions she must answer, on the form she completes, is this:

9. What is the patient's risk of skin infection?

☐

Very High

☐

High

☐

Moderate

☐

Low

Which terms describe the type of data collected by this question?

- (A) Categorical and nominal
- (B) Categorical and ordinal
- (C) Numerical and continuous
- (D) Numerical and discrete
7. Three models of car were road tested, and the fuel consumption rate in litres/100 km was calculated.

Information from the test drive is shown below.

Model	Distance of Test	Fuel Consumed
XR3	150 km	12 litres
MY5	240 km	18 litres
KX2	200 km	15 litres

Which car(s) had the lowest fuel consumption rates?

- (A) The KX2
- (B) The MY5
- (C) The KX2 and XR3 were equal
- (D) The KX2 and MY5 were equal

8. In a game, where two octahedral dice are rolled, the probability of rolling both numbers greater than five is $\frac{9}{64}$.

What is the probability that at least one number is less than six?

- (A) $\frac{9}{64}$
- (B) $\frac{9}{32}$
- (C) $\frac{23}{32}$
- (D) $\frac{55}{64}$
9. Make f the subject of the formula $g = \frac{4}{9}f + 12$.

(A) $f = \frac{9}{4}g - 12$

(B) $f = \frac{9(g+12)}{4}$

(C) $f = \frac{9g-12}{4}$

(D) $f = \frac{9(g-12)}{4}$

10. The heights, in centimetres, of ten students are recorded below.

Tali	Pete	Fatma	Anna	Mia	Keith	Sue	Jack	Gloria	Mick
165	168	169	170	170	171	172	175	179	185

Who would be considered as outliers?

- (A) Mick only
- (B) Mick and Tali
- (C) Tali only
- (D) There are no outliers

Western Mathematics

**2020 YEAR 11 FINAL
EXAMINATION**

Class and Teacher

Mathematics Standard

Student Number

Section II - Answer Booklet

Student Name

70 marks

Attempt Questions 11-36

Allow about 1 hour and 45 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of the booklet.
If you use this space, clearly indicate which question you are answering.

Question 11 (2 marks)

Young's Formula for children's dosages is given below.

2

$$\text{Dosage for children 1 – 12 years} = \frac{\text{age of child (in years)} \times \text{adult dosage}}{\text{age of child (in years)} + 12}$$

A drug called *Meloxiform* has an adult dose of 5 ml.

Use the formula to determine the dose of *Meloxiform* for a nine-year-old child.

(Give your answer correct to the nearest tenth of a millilitre.)

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Question 12 (2 marks)

A scientist measures the length of a bacterial cell to be 0.00000455832 m.

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Write this measurement in scientific notation, correct to 2 significant figures.

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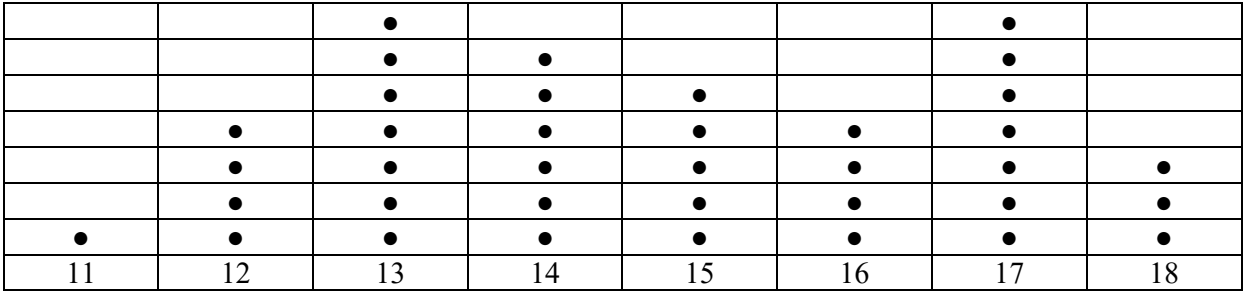
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Question 13 (2 marks)

The dot plot below shows the ages of passengers on a school bus.

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Describe the distribution in terms of shape and modality.



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Question 14 (2 marks)

Use the formula $s = ut + \frac{at^2}{2}$ to find the value of a when $s = 210$, $u = 5$ and $t = 6$.

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Question 15 (2 marks)

Josh bought a bike for \$1 200 and sold it to Mary at a 15% loss a year later.

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Mary fixed the bike up and sold it for \$1 173 a month later.

What percentage profit (or loss) did Mary make on her sale?

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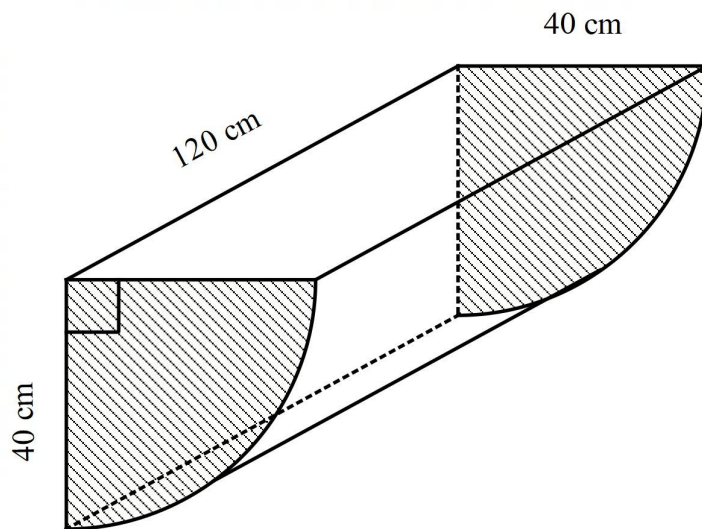
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Question 16 (2 marks)

A food trough at a stable is in the shape of a quarter of a cylinder as shown.

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Given that a volume of $1\,000\text{ cm}^3$ holds one litre, find the capacity, in litres, of the trough.

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Question 17 (2 marks)

At a First Nations conference in NSW, there are 25 different nations represented.

2

The table below shows the probability of meeting members of each of four nations at random at the conference.

Nation	Bundjalung	Dharug	Kamilaroi	Wiradjuri
Probability	$\frac{9}{50}$	$\frac{1}{12}$	$\frac{3}{20}$	$\frac{1}{6}$

There were 600 representatives in total of the 25 nations at the conference.

How many representatives in total were at the conference, from the four nations listed above?

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Question 18 (3 marks)

On the number plane below, draw the graphs of $y = 3x - 3$ and $x + 2y = 8$ and find their point of intersection.

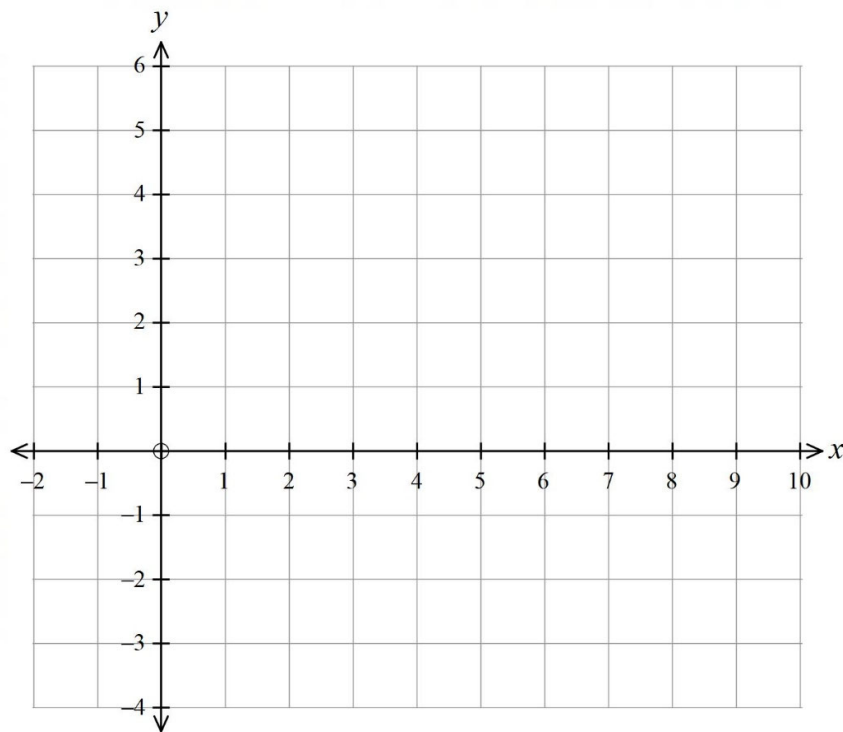
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Question 19 (5 marks)

The times (in minutes) to prepare take-away meals at a café on a particular day are recorded in the grouped frequency distribution table below.

Preparation Time	Class Centre	Frequency	Cumulative Frequency
0 – 2		6	
2 – 4		8	
4 – 6		14	
6 – 8		7	
8 – 10		3	
10 – 12		3	

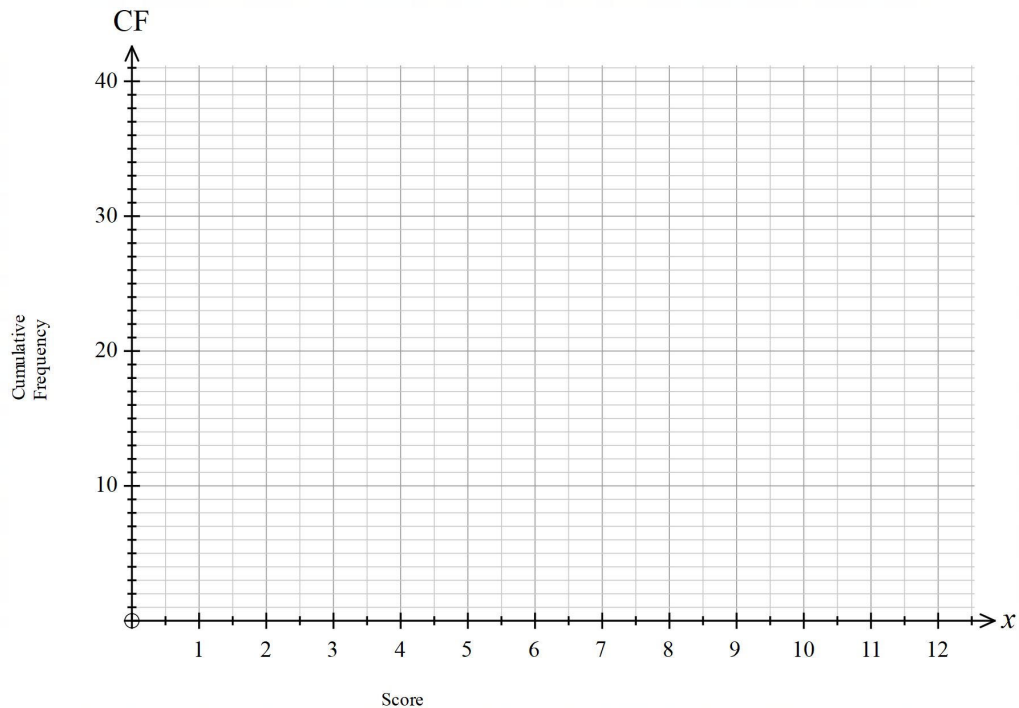
(a) Complete the table above. 2

(b) Use the class centres to calculate the mean of the preparation times, correct to one decimal place. 1

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(c) Complete a cumulative frequency histogram and polygon for the data and estimate the median. 2

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Question 20 (2 marks)

A food manufacturing company produces packets of pasta which have a mass of 250 g.

The packets are boxed in cartons whose mass is 1.2 kg and which hold 30 packets.

For distribution 60 cartons are loaded onto a pallet which has a mass of 25 kg.

Four fully laden pallets are loaded onto a truck.

What is the mass (in tonnes) of the load on the truck?

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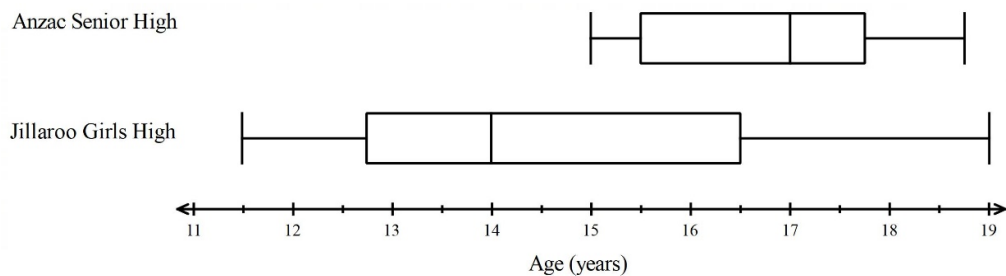
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Question 21 (2 marks)

The box plots below compare the ages of students at two secondary schools.



Using the median and other relevant statistical measures, compare the age profiles of the two schools.

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Question 22 (2 marks)

The table below shows the number of kilojoules burned per kilogram of body weight in an hour for various activities.

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Activity	kJ/kg/h
Cycling	23.0
Jogging	34.1
Rowing machine	33.2
Running	64.5
Swimming	22.1
Walking	18.4
Weight training	35.0

Lily has a body weight of 55 kg and spends two hours on training each day.

One day she spent forty minutes cycling, ten minutes swimming, and the remaining time jogging.

How many kilojoules does she burn in her training session?

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Question 23 (3 marks)

When a passenger vehicle is sold in NSW, stamp duty is payable when the registration is transferred.

The rates are given in the table below.

Value of Vehicle (nearest \$100)	Stamp duty calculation
\$0 – \$45 000	3% of the value of the vehicle
\$45 100 and over	\$1,350 plus 5% of the vehicle's value over \$45,000

Richard intends to buy a car, which is already registered, in a private sale for \$54 200.

- (a) How much is the stamp duty on the transfer of registration?

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- (b) He also insures the car for \$453.00.

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What does it cost him in total to own and insure the car?

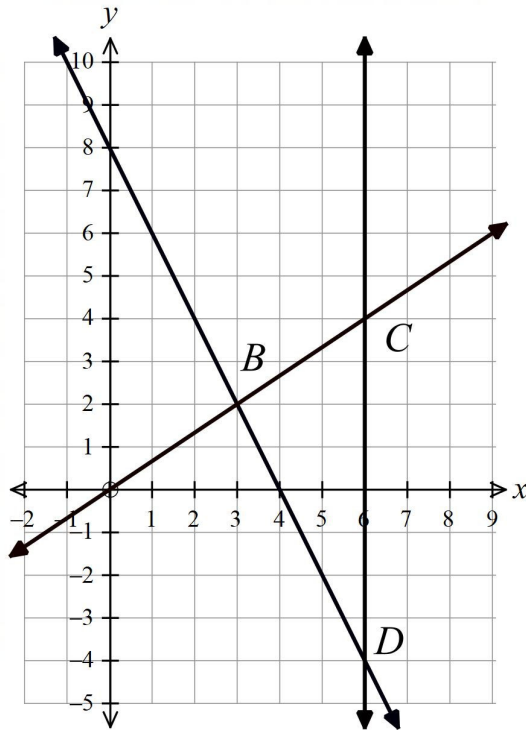
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Question 24 (3 marks)

The graph below shows the points $B(3, 2)$, $C(6, 4)$ and $D(6, -4)$ and three lines joining these points..



(a) What is the equation of the line CD ?

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(b) What is the equation of the line BD ?

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(c) What is the equation of the line BC ?

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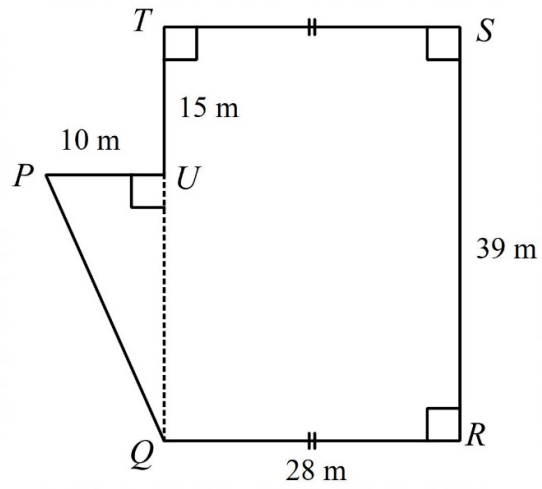
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Question 25 (2 marks)

Calculate the perimeter of the field shown below.

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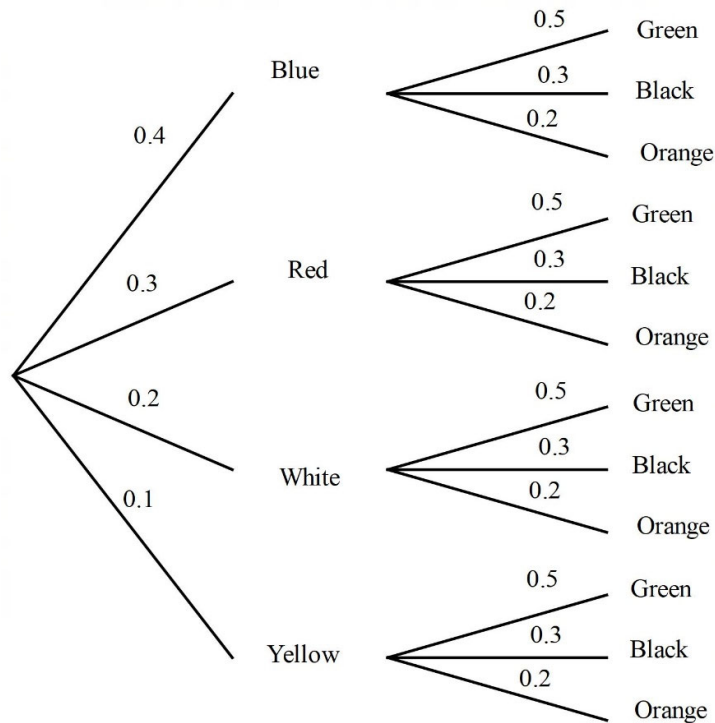
Question 26 (3 marks)

In a board game, players draw one tile from each of two containers.

The first container holds tiles which are red, white, blue and yellow.

The second container holds tiles which are green, black and orange.

The probability tree below shows the likelihood of drawing pairs of coloured tiles at a particular point in the game.



- (a) What is the probability that the player draws a pair of tiles which are blue and black? 1

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- (b) What is the probability that the player does not draw a pair of tiles which are red and orange? 1

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- (c) What is the probability that the player draws a pair of tiles which are either red and black or yellow and green? 1

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Question 27 (2 marks)

Lachie is an author who has eight novels published.

2

In the last financial year, his publisher sold an average of 940 copies of each novel for \$18.95 each.

He earned royalties of 6.25% on the sales of these books.

During the year he was also paid an advance of \$12 500 to write his new novel.

How much did he earn from his writing during the financial year?

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Question 28 (4 marks)

In an Olympic length triathlon, competitors swim three times across a bay which is 500 m wide, then cycle around a 40 km long circuit before running 2 laps of a 5 km long course.

- (a) One competitor takes 30 minutes for the swim leg of the triathlon. 2
What is her swimming speed in km/h?

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- (b) In the bike leg, her normal cycling speed is 35 km/h. 2
How long should the cycle leg take her if she maintains this speed?
(Answer to the nearest minute.)

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Question 29 (2 marks)

A company bought three new computer terminals **three years ago** for \$980 each.

A year ago, it bought another three terminals for \$850 each.

Each of the computers were depreciated using the straight-line method.

The annual loss of value for each terminal is 16% of its initial value.

What is the total value of all the terminals now?

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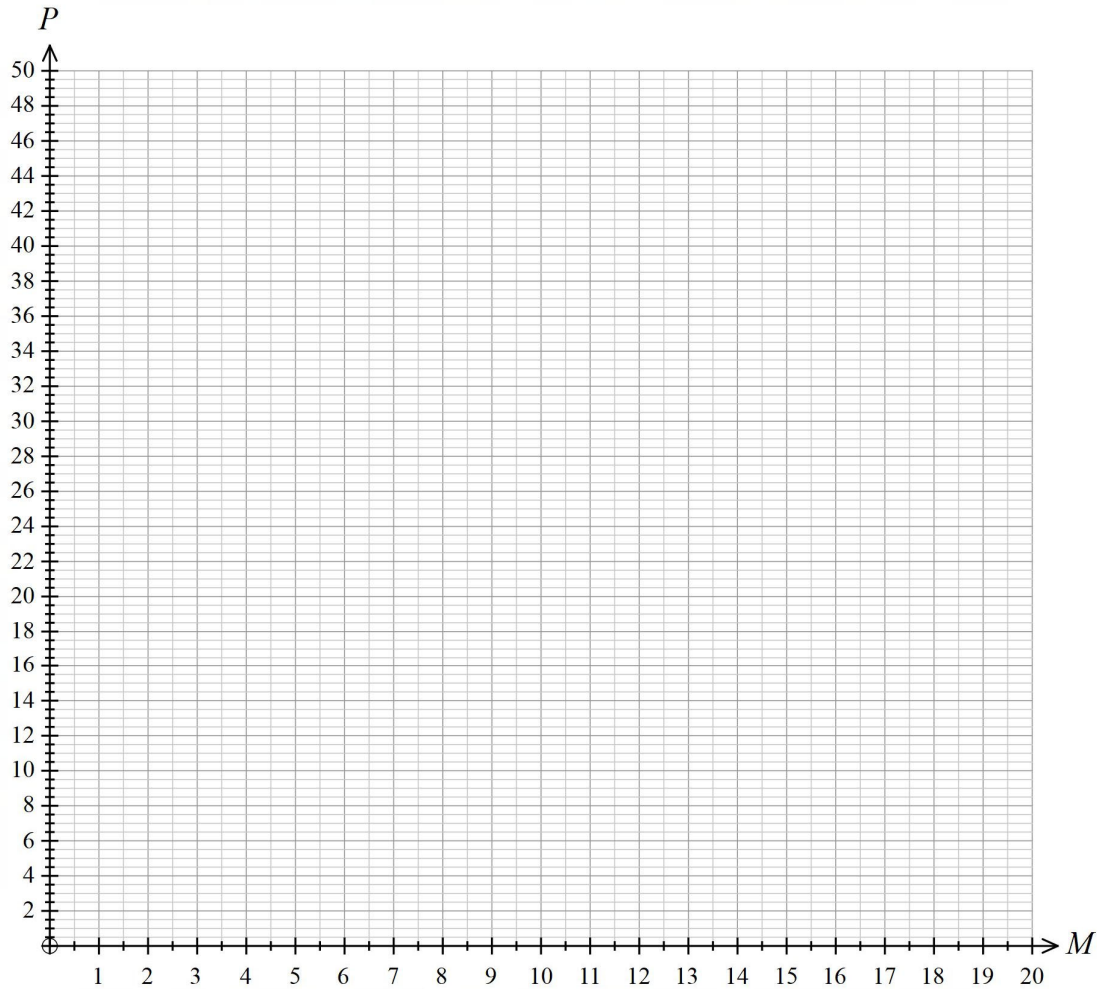
Question 30 (4 marks)

A scientist discovers that for two variables, P varies directly with M .

4

When $M = 12.2$, $P = 30.5$.

Find the equation of the variation, complete the graph of the variation below and hence or otherwise, find the value of P when $M = 18$.



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Question 31 (4 marks)

Connie is paid an annual salary of \$67 626.00 and is paid fortnightly.

She also has investments which provide \$420.00 in interest for the year and income from an investment property worth a total of \$34 000.00.

- (a) Connie has allowable tax deductions totalling \$12 500.00.

1

What is her taxable income for the financial year?

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- (b) Use the table below to find her tax due for the year and determine the amount of her tax refund or tax bill.

3

Income thresholds	Rate	Tax payable on this income
\$0 – \$18,200	0%	Nil
\$18,201 – \$37,000	19%	19c for each \$1 over \$18,200
\$37,001 – \$90,000	32.5%	\$3,572 plus 32.5% of amounts over \$37,000
\$90,001 – \$180,000	37%	\$20,797 plus 37% of amounts over \$90,000

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Question 32 (3 marks)

The owners of a craft store record the material from which the items they sell are made.

The table below gives the frequencies.

Material Used	Frequency	Relative Frequency
Wooden	24	
Metal	16	
Ceramic	28	
Fabric	12	

- (a) Complete the relative frequency column. 1

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- (b) What is the probability that one item, chosen at random will not be made of fabric? 1

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- (c) What is the probability that one item, chosen at random, which is not made of fabric will be made of metal? 1

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Question 33 (2 marks)

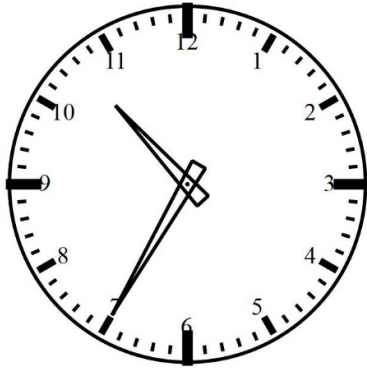
On Monday morning, when Carlo caught a train, his phone was in his bag, so he glanced at the clock on the station platform to check the time.

2

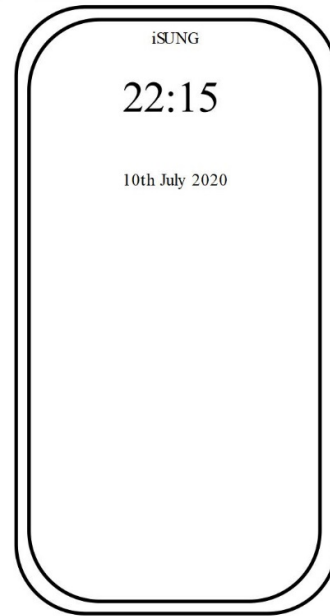
When he arrived at his destination later the same day, he took his phone out and checked the time.

The two times are shown below.

Platform clock on departure



Carlo's phone on arrival



How long did his journey take, in hours and minutes?

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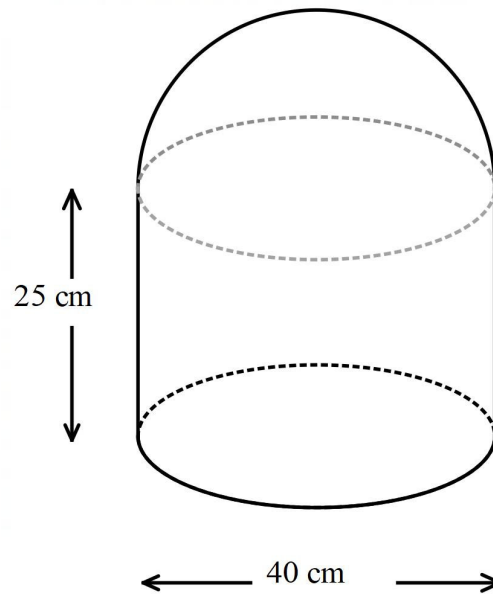
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Question 34 (2 marks)

A designer draws an open glass dome to place over desserts at functions, with the dimensions shown.

2



Calculate the surface area of the outside of the dome, correct to the nearest cm^2 .

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Question 35 (4 marks)

Helen plans to organise an online video conference call between herself and four other participants who are in different countries.

4

The list of participants and their time zones are shown below.

Participant	City	Time zone
Helen	Canberra	UTC+10:00
Kirsty	Glasgow	UTC−0:00
Sitiveni	Suva	UTC+12:00
Joan	Denver	UTC−7:00
Edouard	Pape'ete	UTC−10:00

Helen plans to start the conference at 14:00 on Monday local time in Canberra.

What time (and day) would each of the other participants need to join the conference?

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Question 36 (4 marks)

Part of Andy's gas bill for the three-month period from April to June 2020 is shown below.

Your gas supply details.						Page 2 of 2
Supply address:		6 Elengee Avenue Gastown NSW 2999				
Supply period:		1 Apr 2020 to 30 Jun 2020 (91 days)				
DPI:		52526868				
Energy Plan:		Supersaver				
Meter no. 52526868						
Read date	Read type	Start read	End read	Heating value	Pressure factor	Usage MJ
30 June 20	Actual	3,163	3,267	38.75	1.016	4,094.48
Your next meter read is due 1 Sep 2020 . Please ensure easy access to your meter on these days.						
How we've worked out your bill.						
Previous balance and payments.					Total	
Previous balance				\$183.37		
12 Apr 20 payment				\$183.37cr		
Balance brought forward					\$0.00	
New charges and credits.						
Usage and supply charges		Units	Price	Amount		
Peak		4,094.48MJ	\$0.02967	\$121.48	A	
Supply charge		91 days	\$0.44789	\$40.76	B	
Other charges						
Payment processing fee				\$1.06		
<i>Total charges</i>					+	\$163.30
Credits						
12% Guaranteed Discount				\$19.60cr		
<i>Total credits</i>					-	\$19.60cr
<i>Total new charges and credits</i>					=	\$143.70
<i>Total GST included in new charges and credits</i>					+	\$14.37
Total Due (includes GST)					=	\$158.07

Question 36 continues on page 29

Question 36 continued

- (a) Outline one method that Andy could employ to lower the amounts at **A** and/or **B**, if he wants to reduce his bill.

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- (b) Because Andy used gas for heating in winter, his bill for the months of January to March 2020 was 30% less than this bill.

3

During the months of January to June 2020, as well as gas bills, Andy also had electricity bills which totalled \$910.25.

He anticipates his bills for the months of July to December 2020 will be similar to the total of the bills for January to June 2020.

Andy get his gas and electricity from a single company which has said that it will increase all energy charges by 12% for bills in 2021.

If Andy plans to save a consistent amount each fortnight in 2021 to cover the years bills, how much should he put aside for gas and electricity?

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End of Examination

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

Mathematics Standard 1
Mathematics Standard 2

REFERENCE SHEET

Measurement**Precision**

$$\text{Absolute error} = \frac{1}{2} \times \text{precision}$$

$$\text{Upper bound} = \text{measurement} + \text{absolute error}$$

$$\text{Lower bound} = \text{measurement} - \text{absolute error}$$

Length, area, surface area and volume

$$l = \frac{\theta}{360} \times 2\pi r$$

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(x + y)$$

$$A \approx \frac{h}{2}(d_f + d_l)$$

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV(1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

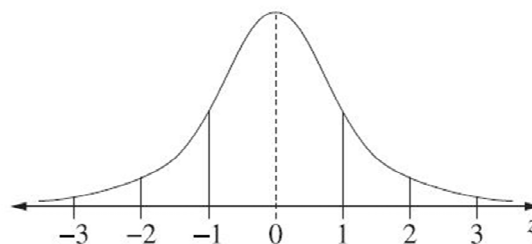
$$z = \frac{x - \bar{x}}{s}$$

An outlier is a score

$$\text{less than } Q_1 - 1.5 \times IQR$$

or

$$\text{more than } Q_3 + 1.5 \times IQR$$

Normal distribution

- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

#

Western Mathematics**2020 Year 11 Final Examination
Mathematics Standard 2**

Name _____ Teacher _____

Section I – Multiple Choice Answer Sheet**Allow about 25 minutes for this section**

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.A ☒ B ☒ ^{correct} C ☐ D ☐

- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |